SECTION 1 · SUBSECTION 1.4
INPUT, NORMALIZATION AND LOWERING

1.4.1 INTRODUCTION.

Input is normalised into one BandishModel, whichever form it arrives in. Lowering then turns the model into a grid the layout can use. VexFlow draws a half note as one glyph, but Bhatkhande notation writes the time out: a note held three beats becomes a swar and two dashes (figure 1.4-1).

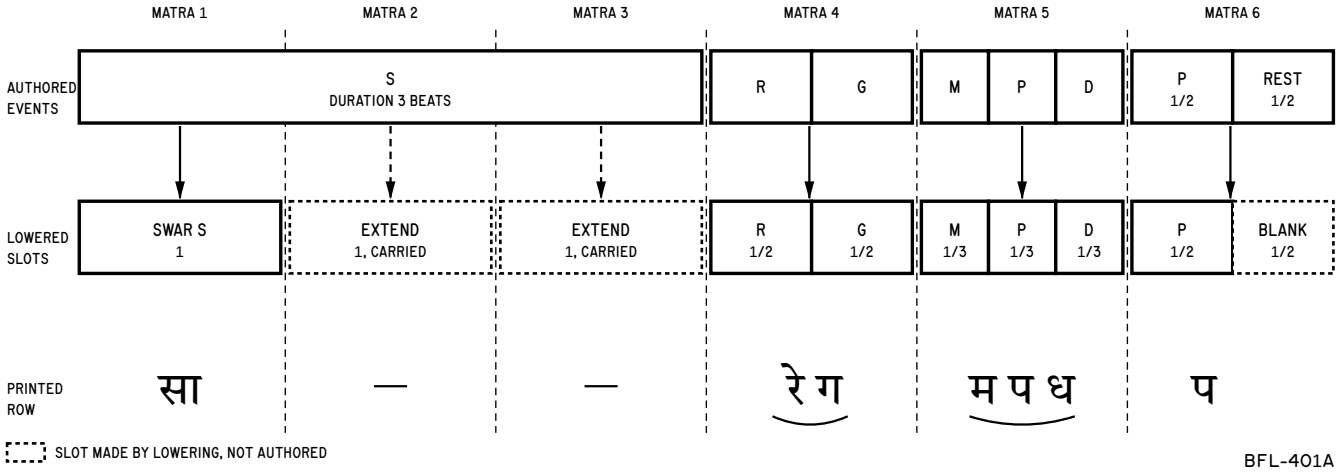


Figure 1.4-1. Lowering Of Durations Into Slots

1.4.2 FUNCTIONAL DESCRIPTION.

lowerBandish() resolves the taal, walks every matra and builds ordered slots whose durations sum to exactly one beat. Events with no duration share the rest of the matra equally. A duration longer than the room left spills into the next matras as carried slots. Taal position, vibhag and marker come from the taal table (figure 1.4-2).

1.4.2.1 Diagnostics.

A musical problem never stops the render. Each one is returned with a level, a code, a message and a path. An error marks inconsistent input; a warning marks input that was padded or ignored.

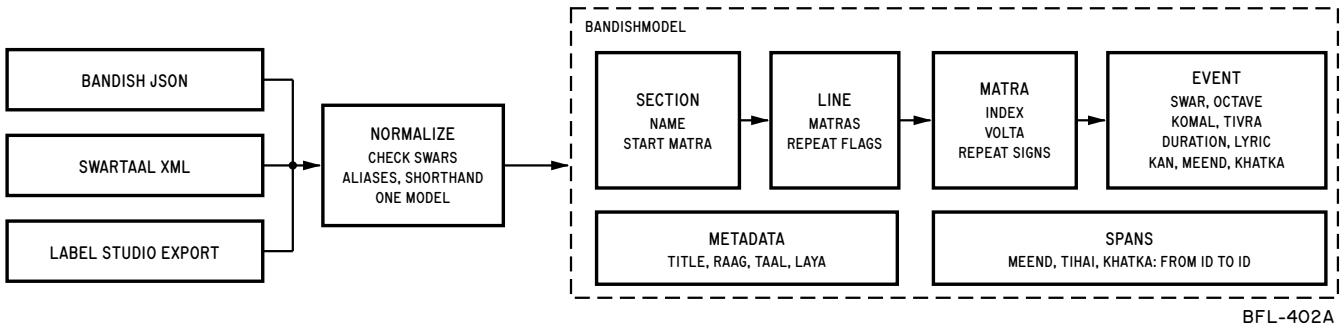


Figure 1.4-2. Input Forms And The Bandish Model

1.4.3 PERFORMANCE AND DESIGN DATA.

Table 1.4-1. Diagnostic Codes

Code	Level and meaning
matra.sequence	WARN: INDEX NOT THE EXPECTED POSITION
matra.underfull	WARN: SHORT OF 1 BEAT, PADDED
duration.overfull	ERROR: DURATIONS ALREADY FILL THE MATRA
carry.overlap	ERROR: HELD NOTE COVERS THE EVENTS
carry.dangling	WARN: SECTION ENDS WHILE HELD
meend.unclosed	WARN: MEEND NEVER CLOSED
tihai.sam	WARN: LAST NOTE MISSES SAM
repeat.offbar	WARN: SIGN NOT ON A VIBHAG BAR
marker.mismatch	WARN: MARKER DIFFERS FROM THE TAAL
matra.overflow	ERROR: EVENT AFTER MATRA IS FULL
rest.blank	INFO: SILENCE PRINTED AS EMPTY SLOT

CAUTION

Bhatkhande has no printed rest sign. A silence is drawn as an empty slot and reported as rest.blank, so a reader cannot tell it from a missing note.

CONVERT AN ANNOTATION USER

```
1 CONVERT - npx tsx
    cli/ls2bandish.ts out.json
    "TITLE" page1.json page2.json
2 RENDER - npm run render --
    out.json out/ls --png
    --no-lyric-curves
  √ compare - render beside scan
```